

WRRRA-Motor H2

From Minimal State Transitions to a Candidate Walking Protein and DNA

Integrated review of the WRRRA Core 1.0 functional architecture, hybrid protein design, and validation boundaries

Researcher	Wonsik Choi
Document	v0.4 Integrated Research Report - English Edition
Date	September 8, 2026
Current status	H2: PREDICTED_SEQUENCE_ONLY Actual walking: OPEN

Central determination

This work maps WRRRA-derived minimum walking requirements onto the established KIF5B motor core and a newly designed 32-aa gate. The protein and coding sequences are computational candidates. Atomic structure, folding, dimeric walking, intracellular cargo transport, and superiority over ordinary coiled coils remain unproven.

Executive Summary

The WRRRA-Motor study begins with a functional question: what is the smallest physical architecture that can repeatedly advance along a polar intracellular track, under thermal noise, while consuming finite chemical free energy? Under the declared strict always-attached contract, applying the WRRRA Core 1.0 principles of minimal computation, fixed-present state, boundary conditions, common carrier, and update yields a minimum functional set: two contacts, non-equilibrium energy input, track polarity, a record embodied in the current physical state, and coordination between the contacts.

Instead of inventing an ungrounded motor from scratch, the protein-level design preserves residues 1-353 of human kinesin-1 KIF5B as an inherited motor core. A designed 32-aa coiled-coil-like WRRRA gate is appended to promote chain association and state coupling, giving the 385-aa hybrid candidate H2-b2f820d18e. Generic human-cell reverse translation gives a 1,158-nt candidate CDS including the stop codon, with 57.69% GC content.

One-sentence conclusion The current result is not a proven artificial walking motor; it is a synthesis-addressable hybrid candidate that maps WRRRA state-transition logic onto a known motor and exposes explicit tests that can falsify the design.

1. Research Question and Scope

1.1 Direct research question

Can WRRRA define the minimum structure of a protein that can physically walk inside a cell and then express that structure as a candidate protein and DNA sequence? Here, walking means repeated plus-end-directed displacement on a polar microtubule, driven by ATP free energy, with one contact retained whenever possible while the other contact is repositioned.

1.2 Included and excluded claims

Layer	Included result	Current status
Functional architecture	Two contacts, drive, polarity, present-state gate, coordination	Derived within model
Hybrid protein	KIF5B 1-353 + designed 32-aa gate	Sequence candidate fixed
DNA	Generic human-cell reverse-translated CDS	Pre-synthesis candidate
Atomic model	Two chains with microtubule and nucleotide states	OPEN_NOT_PREDICTED
Physical function	Folding, dimerization, ATPase, walking, cargo transport	OPEN

2. WRRRA Interpretation of the Walking Problem

2.1 Domain Profile

WRRRA element	Mapping in the walking problem
SOURCE	External chemical free energy, represented by ATP
RELATION / LAW	Binding, unbinding, thermal diffusion, local detailed balance, polar track
STATE / RESIDUE	Head occupancy, nucleotide state, strain, and conformation
BOUNDARY	Cytosol, temperature, thermal noise, load, microtubule lattice and polarity
COMMON CARRIER	Protein conformational change that converts energy state to displacement
UPDATE	ATP-coupled transition of the current physical state
RENDERER	Binding geometry, folding, dimerization, and gate rules
PHENOTYPE	Net displacement toward the microtubule plus end
OBSERVABLE	Velocity, detachment rate, run length, ATP cost, and stall force

Under the WRRRA fixed-present condition, the machine does not keep a separate abstract memory of its history. Information required for the next action must be physically encoded now: ATP or ADP occupancy, which head is bound, and the strain transmitted between the two heads.

2.2 Minimal survival contract

The strict walking contract $V_{\text{strict}}(M)=1$ requires positive expected displacement per cycle, at least one track contact during exchange, finite energetic and structural cost, and a next-step decision determined by the present state. A single-contact Brownian ratchet that fully detaches and rebinds with bias belongs to a separate relaxed contract V_{relaxed} . Therefore, two contacts are not claimed as a universal minimum of nature; they are conditionally minimal under the declared strict contract.

$$V_{\text{strict}}(M) = 1$$

2.3 Conditions for directionality

$$k_+ / k_- = \exp(A), \quad A = (\Delta \mu - W_{\text{load}}) / (k_B T)$$

$$p_+ = 1 / (1 + \exp(-A)), \quad \langle \Delta x \rangle = d \tanh(A/2)$$

If there is no energy drive, or if energy is not coupled to spatial and structural asymmetry, then $A=0$ and the mean displacement is zero. ATP alone does not specify direction. Microtubule polarity and asymmetric protein mechanics must convert chemical free energy into a biased transition cycle.

3. Derivation of the Minimal Architecture

3.1 Conditional minimality of two contacts

On a discrete track, a single point contact moving from site i to site $i+1$ necessarily passes through an interval with zero contacts. It cannot satisfy strict continuous attachment. With two contacts, one may remain as an anchor while the other explores the next site.

$$S0(i,i+1) \rightarrow S1(\text{empty},i+1) \rightarrow S2(i+1,i+2) \rightarrow S0(i+1,i+2)$$

An exhaustive model-level survey of 64 functional architectures combined one to four contacts with four binary conditions: energy drive, polarity coupling, present-state record, and inter-contact coordination. Among strict-contract survivors with two, three, or four contacts, the unique Pareto minimum under $C(M)=(\text{contact count, control count, drive count})$ was the coordinated two-contact architecture.

$$M^* = (2 \text{ contacts, drive, polarity, present record, coordination})$$

3.2 Robustness and redundancy

For a structural sensitivity model, let q be the probability that one contact is lost during an exchange window, and suppose complete detachment of an n -contact architecture requires independent simultaneous loss. Survival over N cycles is then:

$$P_{\text{survive}}(N;n,q) = (1 - q^n)^N$$

Contacts n	$q=0.01$	$q=0.05$	$q=0.10$
1	0.3660	0.0059	0.0000266
2	0.9900	0.7786	0.3660
3	0.9999	0.9876	0.9048
4	0.999999	0.9994	0.9900

These values are not a fit to kinesin kinetics. They show how redundancy broadens the survival range as the boundary becomes harsher. In WRRA terms, extra contacts are not automatically waste; they are a cost paid for robustness under degraded conditions.

3.3 Ideal energy-load bound

$$F d < \Delta \mu, \quad F_{\text{max,ideal}} = \Delta \mu / d$$

Using approximately 20 kBT per ATP at 310 K and an 8-nm step gives an ideal lossless upper bound of about 10.7 pN. This is not a prediction of H2 stall force. It is a thermodynamic ceiling under the stated energy accounting.

4. Post Hoc Comparison with Natural Kinesin

After fixing the WRRRA minimum, comparison with kinesin-1 reveals a direct functional correspondence. The two motor heads provide alternating contacts. The ATPase cycle provides non-equilibrium drive. The microtubule supplies track polarity. Nucleotide occupancy and neck-linker strain encode the present state. Mechanical coupling between the heads biases the order of binding and release.

WRRRA requirement	Natural kinesin analogue	H2 implementation
Two contacts	Two kinesin motor heads	Two copies of KIF5B 1-353
Energy drive	ATP hydrolysis cycle	Inherited motor-domain chemistry
Polar track	Microtubule lattice	Inherited binding interface
Present-state record	Nucleotide state and strain	Inherited core plus designed gate hypothesis
Coordination	Head-head gating	Gate intended to transmit and constrain strain

Interpretation boundary This correspondence does not prove that WRRRA independently rediscovered every kinesin mechanism or that the H2 gate implements natural head-head gating. It identifies a coherent mapping between the abstract contract and a known physical motor.

5. Candidate Lineage and Design Strategy

5.1 Role of H1

H1 was a calibration construct based on KIF5B 1-353 plus a known GCN4-type dimerization segment. It established that the renderer could preserve an inherited motor core, append a dimerization region, label ownership, and close the sequence and DNA bookkeeping. H1 was not the differentiated WRRRA design.

5.2 Why H2 was selected

H2 preserves the same 353-aa KIF5B core and replaces the calibration tail with a designed 32-aa gate: VQNIEQKIANLKEEGAAALQQVEQKIQNLKAE. The gate was selected by a constrained renderer for coiled-coil-compatible periodicity, charge pattern, length, and absence of obvious sequence-level forbidden motifs. Only 8.31% of the 385-aa chain is newly designed; 91.69% is inherited.

5.3 Why N1 was not emitted

A fully de novo autonomous motor N1 would require a validated renderer capable of jointly predicting an ATPase fold, microtubule interface, mechanically coupled multistate cycle, dimer topology, and negative-design landscape. That validator does not exist in the present work. WRRRA therefore applies fail-closed behavior: it does not emit a plausible-looking but unsupported N1 sequence.

6. Protein Renderer 0.3

Protein Renderer 0.3 is a constraint-and-provenance engine, not a general protein foundation model. It accepts inherited segments, a gate design space, topology assumptions, sequence constraints, and explicit failure conditions. Candidate ranking uses a heuristic score to order survivors; it does not estimate the probability that a protein will walk.

Renderer check	H2 result	Meaning
Total chain length	385 aa	Within declared construction range
Inherited fraction	91.69%	Known motor core retained
Designed fraction	8.31%	Novelty limited to gate
Gate syntax	Pass	Periodic and charge-pattern constraints satisfied
Switch proxy	1.000	Sequence-level proxy only; not kinetics
Rank score	5.0806	Heuristic constraint score; not success probability
Structure coordinates	Not generated	OPEN_NOT_PREDICTED

7. Detailed Architecture of H2-b2f820d18e

Residues	Ownership	Assigned role	Evidence boundary
1-353	INHERITED_KIF5B	ATP motor core, microtubule interface, neck-linker region	Known source segment; function in this truncation still must be tested
354-385	DESIGNED_GATE	Parallel association and state/strain coupling hypothesis	Sequence constraints passed; physical role unverified
Two chains	PREDICTED_TOPOLOGY	Parallel homodimer	No atomic model or oligomer measurement

A key risk is the 1-353 boundary. A natural or commonly used active kinesin construct may include additional stalk or engineered dimerization sequence. The designed gate may fail to replace GCN4-like stabilization while preserving the neck-linker strain geometry needed for processive stepping. Thus, preserving the motor-domain amino acids is not equivalent to preserving the full mechanochemical cycle.

8. DNA Renderer Result

Property	Value
Protein length	385 aa
CDS length including stop	1,158 nt
Coding bases excluding stop	1,155 nt
GC content	57.69%
Translation check	Exact match to candidate protein
Status	PREDICTED_NOT_SYNTHESIS_READY

The DNA is a generic human-cell reverse translation of the candidate protein. It is not a complete expression cassette. No promoter, Kozak context, untranslated regions, intron design, polyadenylation signal, vector backbone, selectable marker, purification tag, localization signal, cargo-binding domain, or biosafety containment strategy is specified.

Length accounting 385 amino acids require 385 codons = 1,155 coding nucleotides. Adding one 3-nt stop codon gives a 1,158-nt CDS. Regulatory and vector elements would make the full construct substantially longer.

9. Predicted Walking Cycle

The animation and figure are a visualization of a state machine. They are not molecular dynamics, single-molecule tracking, or evidence that H2 physically moves.

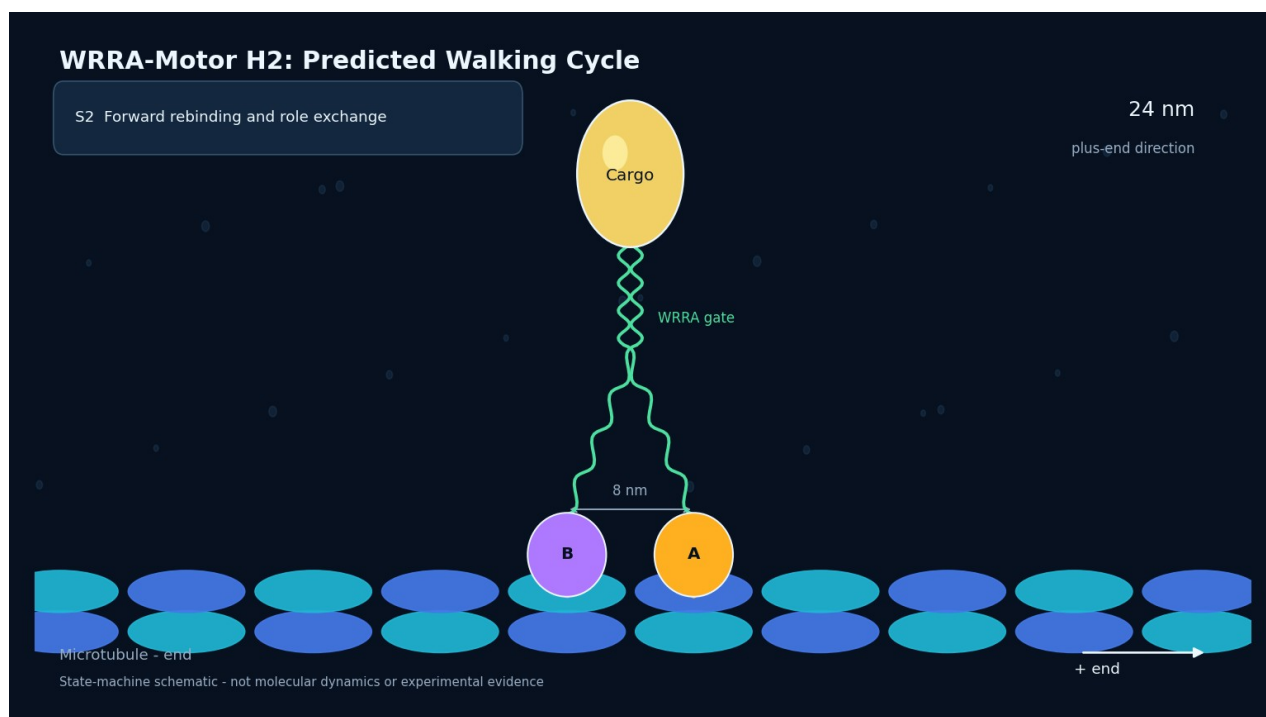


Figure 1. Predicted H2 walking cycle. Schematic only; not an experimental trajectory or molecular-dynamics simulation.

9.1 State vector

$$S = (x_A, x_B, b_A, b_B, n_A, n_B, \epsilon)$$

The state records the two head positions x , binding states b , nucleotide states n , and an effective strain or gating variable ϵ . The designed gate is hypothesized to influence ϵ and the allowed transition order; no physical parameterization has yet been calculated.

9.2 S0 - both contacts bound

Both heads occupy adjacent microtubule binding sites. One is designated rear and the other front. The two-chain structure supplies the anchor-and-explorer geometry required by the strict contract.

9.3 S1 - ATP input and rear-head movement

ATP-coupled conformational change is assumed to bias release of the rear head while the front head remains attached. The released head diffuses through a constrained forward search. The gate is intended to reduce simultaneous detachment, maintain asymmetric strain, and suppress transitions that violate the cycle. These roles remain design intentions, not measured properties.

9.4 S2 - forward rebinding and role exchange

The moving head binds the next site. Two-contact attachment is restored and the head roles exchange. The internal state returns to an equivalent form translated by one microtubule repeat.

$$S2(A,B) \text{ equivalent to } S0(B,A) + 8 \text{ nm}$$

$$x_n = x_0 + 8n \text{ nm}$$

10. Differentiating Features of the Research

10.1 The design logic precedes the sequence

WRRA-Motor first defines the state cycle and survival contract, converts each transition into physical requirements, then maps those requirements into topology, amino-acid sequence, and DNA. The differentiator is therefore the traceable derivation, not merely the existence of another kinesin fusion sequence.

State cycle -> physical conditions -> minimal architecture -> protein topology -> amino acids
-> DNA

10.2 Preserve validated function and minimize change

The ATP motor and microtubule-binding surface are not reinvented. They are inherited from KIF5B. The designed region is limited to the 32-aa gate. This operationalizes the WRRA principle of preserving stable present structure and changing only what is required by the missing transition.

10.3 An auditable line from abstraction to synthesis

The work connects WRRA principles, a state machine, a mechanistic hypothesis, protein topology, a 385-aa candidate, and a 1,158-nt CDS in one provenance ledger. Every segment and conclusion is marked INHERITED, DESIGNED, PREDICTED, OPEN, or FAIL-CLOSED.

10.4 Condition for a genuine differentiated claim

Critical test WRRA-specific differentiation becomes supportable only if H2 produces a reproducible measurable behavior that is superior to or distinct from a length-matched generic coiled coil, a composition-preserving scrambled gate, and established kinesin controls.

11. How Difficult Is This Level of Protein Design?

Stage	Difficulty	H2 status
Write amino-acid and DNA candidates	Relatively easy	Completed
Codon and forbidden-motif checks	Relatively easy	Completed
Static structure prediction	Moderate	Not run
Dimer + nucleotide + microtubule complex	Hard	Not run
Multistate mechanochemical dynamics	Very hard	Not run
Processive single-molecule walking	Specialized experiment	OPEN
Selective intracellular cargo transport	Very hard	Design incomplete

Stage	Difficulty	H2 status
Fully de novo autonomous motor	Research frontier	N1 fail-closed

It is now comparatively easy to draft a sequence, reverse-translate it, and run syntax-level screens. It remains difficult to establish that the sequence folds, couples ATP hydrolysis to directional stepping, stays processive under load, and functions safely in a cell. H2 is more tractable than a fully de novo motor because it reuses a natural core, but the gate can still disrupt rather than improve function.

12. What Can Be Proved Without Experiments?

12.1 Formal result: a conditional state-machine theorem

If every normal cycle implements $S0(x) \rightarrow S0(x+8 \text{ nm})$ and backward cycles are excluded by definition, then induction proves $x_n = x_0 + 8n \text{ nm}$ after n cycles. This proves directional accumulation inside the declared transition system.

$$S0(x) \rightarrow S1(x) \rightarrow S2(x+8 \text{ nm}) \rightarrow S0(x+8 \text{ nm})$$

Exact meaning of the proof If the WRRRA transition conditions are physically implemented, the state machine advances. The proof does not establish that this particular 385-aa sequence implements those conditions.

12.2 Evidence that computation can strengthen

Pre-experimental computation can test whether H2 plausibly folds, forms a parallel homodimer, preserves the ATP pocket and microtubule interface, retains nucleotide-dependent structural differences, transmits strain through the neck-linker/gate junction, and avoids dominant aggregation or off-target states. Such results remain conditional on the predictor, force field, initial conditions, and sampling depth.

$$\text{Computational evidence} = P(\text{function} \mid \text{model, force field, initial conditions, sampling})$$

12.3 Claims that still require experiments

Claim	Required evidence
H2 is expressed and soluble in cells or in vitro	Expression and biochemical measurements
Two 385-aa chains form the intended dimer	Oligomer-state measurement
ATPase activity is retained	Enzymatic assay
The motor walks toward the microtubule plus end	Single-molecule motility assay
The gate improves coordination over generic coiled coil	Controlled comparison

Claim	Required evidence
The construct transports a selected intracellular cargo	Additional cargo design and cell experiment

13. Validation Roadmap

13.1 Computational validation

Stage	Question	Example pass criterion
A. Monomer	Are the KIF5B core and ATP pocket preserved?	High-confidence core; no junction clash
B. Dimer	Does the gate form a parallel homodimer?	Consistent two-chain interface
C. Multistate	Are ATP, ADP, and apo states distinguishable?	Neck-linker state differences retained
D. Complex	Is the microtubule interface preserved?	Known binding surface and polarity retained
E. Dynamics	Does the gate transmit strain without collapse?	Stability across repeated samples
F. Negative design	Do aggregation or dead states dominate?	High-risk candidates eliminated

13.2 Experimental comparison set

Group	Construct	Purpose
P0	Active KIF5B 1-420 or 1-560 class motor	Positive assay control
B0	KIF5B 1-353	No-gate baseline
H1	1-353 + GCN4	Established dimerization calibration
H2	1-353 + WRRA gate	Primary test candidate
C1	1-353 + length-matched generic coiled coil	Separate generic dimerization effect
C2	1-353 + composition-preserving scrambled gate	Test sequence-order specificity

Primary observables should include expression and solubility, oligomer state, ATPase rate, microtubule landing rate, velocity, run length, detachment rate, simultaneous-head detachment, and load response. H2 must recover activity relative to B0 and show a reproducible distinction from H1, C1, and C2 before a WRRA-gate-specific contribution can be claimed.

14. Claim and Provenance Ledger

Item	Ownership / grade	Permitted statement
WRRRA minimum contract	DERIVED / model-internal	Coordinated two-contact form is Pareto-minimal under the strict contract
KIF5B 1-353	INHERITED	An established motor-core sequence is used
Gate 354-385	DESIGNED	A candidate satisfying declared sequence grammar
H2 topology	HYBRID / PREDICTED	Parallel homodimer hypothesis
DNA 1,158 nt	PREDICTED	Generic reverse-translated CDS candidate
Atomic structure	OPEN	Coordinates not predicted
Physical walking	OPEN	Not experimentally demonstrated
Animation / figure	SCHEMATIC	Predicted state-machine visualization
Fully de novo N1	FAIL-CLOSED	No sequence emitted without a validator

15. Risks and Failure Interpretation

Risk	Possible outcome	Interpretation
Incorrect gate oligomerization	Monomer or higher oligomer instead of parallel dimer	Sequence grammar passed, topology failed
Unstable GAAA hinge	Flexible linker or collapse point rather than switch	Limit of switch proxy
Junction perturbs strain	ATPase retained but processivity reduced	Core-gate integration failed
Head cycles do not coordinate	Both heads bind, detach, or stall together	State-coupling hypothesis failed
Excess activity	Transport disruption or toxicity	Functional success separated from safety
No difference from controls	Same behavior as generic coiled coil	WRRRA-specific contribution unsupported

16. Conclusion

WRRR-Motor applies a minimum-computation framework to intracellular walking and derives a functional architecture consisting of two contacts, energy drive, polarity, present-state gating, and coordination. It then preserves a KIF5B motor core, introduces only a 32-aa designed gate, and connects the 385-aa hybrid candidate to a 1,158-nt DNA candidate.

The strongest current result is not proof of a new walking protein. It is a reproducible and auditable design lineage from functional conditions to topology, sequence, DNA, and a falsification plan, combined with fail-closed handling of an unsupported fully de novo motor.

Without experiments, the work can prove only a conditional theorem about the declared state transitions. Structure prediction and simulation may increase or decrease confidence, but expression, dimerization, ATPase activity, single-molecule motility, and comparison against controls are required to establish physical walking and any genuine WRRR-specific advantage.

Accurate present claim H2 is a synthesis-addressable hybrid motor candidate that preserves a known kinesin motor segment and attempts to implement WRRR-required state coordination with a 32-aa gate. Repeated forward motion is valid within the model; actual folding and walking remain OPEN.

Appendix A. Top H2 Amino-Acid Sequence

Candidate ID: H2-b2f820d18e | Total: 385 aa | Residues 1-353: INHERITED_KIF5B | Residues 354-385: DESIGNED_GATE

MADLAECNIK VMCRFRPLNE SEVNRGDKYI AKFQGEDTVV IASKPYAFDR VFQSSTSSEQ
VYNDCAKKIV KDVLEGYNGT IFAYGQTSSG KTHTMEGKLH DPEGMGIIPR IVQDIFNYIY
SMDENLEFHI KVSYFEIYLD KIRDLLDVSF TNLVHEDKN RVPYVKGCTE RFVCSPDEV
DTIDEGKSNR HVAVTNMNEH SSRSHSIFLI NVKQENTQTE QKLSGKLYLV DLAGSEKVS
TGAEGAVLDE AKNINKSLSA LGNVISALAE GSTYVPYRDS KMTRILQDSL GGNCRTTIVI
CCSPSSYNES ETKSTLLFGQ RAKTIKNTVC VNVELTAEQW KKKYEKEKEK NKIVQNIEQ
IANLKEEGAA ALQQVEQKIQ NLKAE

Region	Length	Sequence / note
Inherited core	353 aa	Exact first 353 residues shown above
Designed gate	32 aa	VQNIEQKIANLKEEGAAALQQVEQKIQNLKAE

Sequence integrity check: 385 aa; alphabet valid = True.

Appendix B. Candidate Coding DNA Sequence

Status: PREDICTED_NOT_SYNTHESIS_READY | Generic human-cell reverse translation | Stop codon included

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ATGGCAGACC TCGCAGAATG CAACATCAAG GTCATGTGCC GCTTCCGCCC CTTGAACGAG
AGCGAGGTCA ACCGCGGAGA TAAGTACATC GCCAAGTTCC AGGGCGAGGA CACCGTCGTC
ATCGCAAGCA AGCCTTACGC CTTGACAGA GTCTTCCAGA GCAGCACCAG CCAGGAGCAA
GTCTACAACG ACTGCGCCAA GAAGATCGTC AAGGACGTCC TCGAGGGCTA CAACGGCACC
ATCTTCGCCT ACGGCCAGAC AAGCAGCGGA AAGACCCACA CCATGGAGGG AAAGCTCCAC
GACCCTGAAG GAATGGGAAT CATCCCAGA ATCGTCCAGG ACATCTTCAA CTACATCTAC
AGCATGGACG AGAACCTCGA GTTCCACATC AAGGTCAGCT ACTTCGAGAT CTACCTCGAC
AAGATCCGCG ACCTCCTCGA CGTCAGCAAG ACCAACCTCA GCGTCCACGA GGACAAGAAC
CGCGTCCCCT ACGTCAAGGG CTGCACCGAG CGCTTCGTCT GCAGCCCCGA CGAGGTCATG
GACACCATCG ACGAGGGCAA GAGCAACGCG CACGTCGCCG TCACAAACAT GAACGAGCAC
AGCAGCCGCA GCCACTCTAT CTTCTCATC AACGTCAAGC AGGAGAACAC CCAGACCGAG
CAGAAAGCTCA GCGGCAAGCT CTACCTCGTC GACCTCGCCG GAAGCGAGAA GGTGAGCAAG
ACCGGAGCAG AGGGAGCAGT TCTCGATGAG GCAAAGAACA TCAACAAGAG CCTCAGCGCC
CTCGGCAACG TCATCAGCGC CTTGGCAGAG GGATCTACCT ACGTCCCTTA CCGCGATAGC
AAGATGACCC GCATCCTCCA GGACAGCCTC GGAGGAAACT GCAGAACCAC CATCGTCATC
TGCTGCAGCC CTAGCTCTTA CAACGAGAGC GAGACCAAGA GCACCTCCTT CTTGCGCCAG
AGAGCAAAGA CCATCAAGAA CACCGTCTGC GTCAACGTCG AGCTCACC GCAGCAGTGG
AAGAAGAGT ACGAGAAGGA GAAGGAGAAG AACAAATCG TCCAGAACAT CGAGCAGAA
ATCGCCAACC TCAAGGAGGA GGGCGCCGCC GCCCTCCAGC AGGTCGAGCA GAAGATCCAG
AACCTCAAGG CCGAGTAA

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DNA integrity check: 1158 nt including stop | Encoded amino acids: 385 | GC: 57.69% | Terminal codon: TAA.

Appendix C. Reference and Evidence Notes

1. Hua W et al. Coordinated movement of kinesin motor domains determined by single-molecule imaging. *Nature* (1997).
2. Coy DL et al. Kinesin's tail domain is an inhibitory regulator of the motor domain. *Journal of Biological Chemistry* (1999).
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8. Nilsson P et al. A synthetic molecular motor moves on a DNA track. *Nature Nanotechnology* (2026). DOI: 10.1038/s41565-026-02211-3.
9. UniProt P33176: Kinesin-1 heavy chain KIF5B, Homo sapiens.
10. RCSB Protein Data Bank entry 1MKJ: kinesin motor-domain structural reference.
11. Addgene plasmid #193714: KIF5B motor construct reference.

Reference note: the cited literature provides context for kinesin mechanism, structural modeling, and molecular motor design. It does not constitute validation of H2. H2-specific claims must be supported by H2-specific computational and experimental evidence.